FUNCTIONS

WORKED OUT EXAMPLES

W.E.1: The function $f(x)=|px-q|+r|x|$, where $x\in(-\infty,\infty)$ where $p>0, q>0, r>0$ assume its minimum value only at one point, if

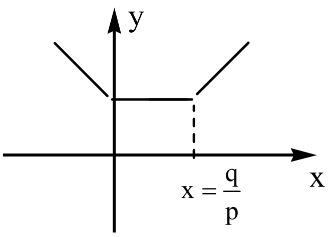
a) $p\neq q$ b) $r\neq q$ c) $r\neq p$ d) $p=q=r$

Key. c

Sol. $f(x)=\begin{cases}-px+q-rx; & x\leq 0\\ -px+q+rx; & 0<x\leq \dfrac{q}{p}\\ px-q+rx; & \dfrac{q}{p}<x\end{cases}$

(i) for $r=p$, $f'(x)\begin{cases}<0; \ if \ x<0\\ =0; \ if \ 0<x<\dfrac{q}{p}\\ >0; \ if \ x>\dfrac{q}{p}\end{cases}$

Here, we have infinitely many point for m in value for $f(x)$.

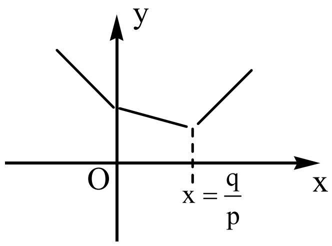


(ii) for $r<p$, $f'(x)<0; \ if \ x\leq 0$

$<0; \ if \ 0<x\leq \dfrac{q}{p}$

$>0; \ if \ x>\dfrac{q}{p}$

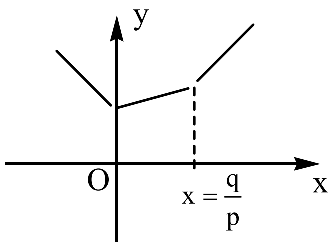
Here we have only one point of min. of $f(x)$ at $x=\dfrac{q}{p}$



(iii) for $r>p$, $f(x)<0; \ if \ x\leq 0$

$>0; \ if \ x>0$

form the graph, we have only one point of minimum at $x=0$.



W.E-2: If $f(x)=x^2+2bx+2c^2$ and $g(x)=-x^2-2cx+b^2$ such that min $f(x)>$ max $g(x)$, then the relation between 'b' and 'c' is

a) no real value of b & c

b) $0<c<b\sqrt{2}$ c) $|c|<|b|\sqrt{2}$

d) $|c|>b\sqrt{2}$

Key. d

Sol. We have min. $f(x)=2c^2-b^2$;

max. $g(x)=b^2+c^2$

$f_{\min}>g_{\max}\Rightarrow 2c^2-b^2>b^2+c^2$

$\Rightarrow |c|>|b|\sqrt{2}$

W.E-3: X and Y are two sets and $f: X \to Y$. If $\{f(c) = y; c \subset X, y \subset Y\}$ and $\{f^{-1}(d) = x; d \subset Y, x \subset X\}$, then the true statement, is

a) $f\left(f^{-1}(b)\right) = b$ b) $f^{-1}\left(f(a)\right) = a$

c) $f\left(f^{-1}(b)\right) = b;\ b \subset y$

d) $f^{-1}\left(f(a)\right) = a;\ a \subset x$

Key. d

Sol. Given that X and Y are two sets and $f: X \to Y$, $\{f(c) = y; c \subset X, y \subset Y\}$ and $\{f^{-1}(d) = x; d \subset Y, x \subset X\}$

since $f^{-1}(d) = x \Rightarrow f(x) = d$

Now if $a \subset X \Rightarrow f(a) \subset f(X) = d$

$$\Rightarrow f^{-1}f(a) = a$$

$f^{-1}f(a) = a,\ a \subset X$

W.E-4: If $f(x)$ satisfies the functional equation

$$x^2.f(x) + f(1-x) = 2x - x^4, \text{ then } f\left(\frac{1}{3}\right) =$$

a) $\frac{1}{3}$ b) $\frac{1}{9}$ c) $\frac{8}{9}$ d) $\frac{10}{9}$

Key. c

Sol. Replace 'x' by '$1-x$', we have

$$(1-x)^2.f(1-x) + f(x) = 2(1-x) - (1-x)^4$$

Now eliminating $f(1-x)$

$$\Rightarrow f(x) = \frac{\left(2x - x^4\right)(1-x)^2 - 2(1-x)(1-x)^4}{x^2(1-x)^2 - 1}$$

$$= 1 - x^2 \Rightarrow f\left(\frac{1}{3}\right) = \frac{8}{9}$$

W.E-5: Let $f: R \to R$ be a function such that for some $p > 0$, $f(x+p) = \frac{f(x)-5}{f(x)-3}$, $\forall x \in R$, then the period of 'f' is

a) 2p b) 4p c) 3p d) 5p

Key. b

Sol. 3 does not belong to the range of 'f'

Now

$$f(x+2p) = f(x) \Rightarrow f(x+p+p) = f(x)$$

$$\Rightarrow \frac{\frac{f(x)-5}{f(x)-3} - 5}{\frac{f(x)-5}{f(x)-3} - 3} = \frac{-4f(x)+10}{-2f(x)+4}$$

$$= \frac{2f(x)-5}{f(x)-2}$$

$(f(x)-2)^2 = -1$, which is absurd

$\Rightarrow f(x+2p) \neq f(x)$

Similarly $f(x+3p) \neq f(x)$

Now

$$f(x+4p) = \frac{f(x+3p)-5}{f(x+3p)-3} = \frac{\frac{3f(x)-5}{f(x)-1} - 5}{\frac{3f(x)-5}{f(x)-1} - 3} = f(x)$$

$\Rightarrow$ Period of $f(x)$ is 4p

W.E-6: For any real number 't', [t] denotes the G.I.F. Let W be the set of all non-negative integers and $f: W \to W$ is a function defined by

$$f(x) = \left(x - 10.\left[\frac{x}{10}\right]\right).10^{[\log_{10} x]} + f\left(\left[\frac{x}{10}\right]\right); x > 0$$

$$= 0; \qquad x = 0$$

if $f(7752)$ is a four digit number abcd then $(c+d)-(a+b) =$

a) 5 b) 7 c) 3 d) 2

Key. b

Sol. $f(7752)=2557 \Rightarrow c+d-(a+b)=7$

W.E-7: Suppose $f:[-2,2]\to R$ defined by $f(x)=\begin{cases}-1, & for\ -2\le x<0\\ x-1; & for\ 0\le x\le 2\end{cases}$ then

$\{x \mid x\in[-2,2]: x\le 0\ and\ f(|x|)=x\}=$

a) $\{-1\}$ b) $\{0\}$ c) $\left\{\frac{-1}{2}\right\}$ d) ϕ

Key. c

Sol. $|x|\in[0,2]$

$\therefore f(|x|)=|x|-1=x \Rightarrow x=-\frac{1}{2} \quad (x\le 0)$

W.E-8: Let $f:\left(\frac{1}{2},1\right)\to[0,\infty)$ and

$f(x)=\sqrt{\log_e\left(\frac{x}{1-x}\right)}$ then $f(x)$ is

a) one-one function b) Many-one function
c) Into function d) bijection

Key. c

Sol. Range is $(0,\infty)$

W.E-9: Number of positive integers in the domain of the function $f(x)=\sqrt{\log_{0.5}\log_6\left(\frac{x^2+x}{x+4}\right)}$ is

Key. 6

Sol. Integers are 3, 4, 5, 6, 7, 8

W.E-10: Let $f(x)$ be a non-constant polynomial satisfying the relation

$f(x).f(y)=f(x)+f(y)+f(xy)-2$, for all real '$x$', '$y$' and $f(0)\ne 1$, suppose $f(4)=65$, then

a) $f'(x)$ is a polynomial of degree 2

b) roots of $f'(x)=2x+1$ are real

c) $x.f'(x)=3(f(x)-1)$

d) $f'(-1)=3$

Key. a, b, c, d

Sol. Given

$f(x).f(y)=f(x)+f(y)+f(xy)-2,\ \forall x,y\in R$

Put $x=1=y$ then

$(f(1))^2-3f(1)+2=0 \Rightarrow f(1)=1\ (or)\ 2$

If $f(1)=1$ then $f(1)=1, \forall x\in R$. A contradiction

$\because$ degree of $f(x)$ is positive

$\therefore f(1)\ne 1$. Hence $f(1)=2$

replace 'y' by '$\frac{1}{x}$' then $f(x).f\left(\frac{1}{x}\right)$

$(\because f(1)=2)$

$f(x)=\pm x^n+1 \Rightarrow f(4)=65$

$\Rightarrow f(x)=x^3+1 \Rightarrow f'(x)=3x^2$

W.E-11: The least value of the function

$f(x)=\left(\frac{2+\cos x}{\sin x}\right)$ for $0<x<\pi$, is

a) 2 b) $\frac{1}{2}$ c) 3 d) $\frac{1}{3}$

Key. c

Sol. Put $\cos x=t$ sothat $-1<t<1$

$f(t)=\frac{(2+t)^2}{1-t^2} \Rightarrow f'(t)=0 \Rightarrow t=-\frac{1}{2}$

$f'(t)<0$ for $t<-\frac{1}{2}$; $f'(t)>0$ for $t>-\frac{1}{2}$

$\Rightarrow f(t)$ is least at $t=-\frac{1}{2}$ and $f(t)=3$

W.E-12: The range of the function

$$f(x)=\log_2\left(2-\log_{\sqrt{2}}\left(16\sin^2 x+1\right)\right)$$

a) $[0,1]$ b) $(-\infty,1]$ c) $[-1,1]$ d) $(-\infty,\infty)$

Key. b

Sol. Here $16\sin^2 x+1>0$ and

$$2-\log_{\sqrt{2}}\left(16\sin^2 x+1\right)>0$$

W.E-13: The range of $f:R\to R,$

$$f(x)=\frac{\left(\sqrt{x^2+1}-3x\right)}{\sqrt{x^2+1}+x}$$ is

a) $(0,\infty)$ b) $(-1,\infty)$

c) $(-\infty,-1)$ d) $[1,\infty)$

Key. b

Sol. Let $\sqrt{x^2+1}+x=t\Rightarrow\sqrt{x^2+1}-x=\frac{1}{t}$

$$\Rightarrow f(t)=-1+\frac{1}{t^2}\qquad\Rightarrow \text{range}=(-1,\infty)$$

W.E-14: If $f(x)+f(y)=f(x+y)f(x-y)\ \forall x,y$

then $f(x)$ is

a) even b) odd

c) Neither even nor odd

d) both even, odd

Key. a

Sol. Interchange x and y

$$\Rightarrow f(y)+f(x)=f(y+x)f(y-x)$$

$$\Rightarrow f(x-y)=f(y-x)\quad \text{putting } y=2x$$

$$\Rightarrow f(x)=f(-x)$$

W.E-15: The number of roots of the equation $|\cos|x||=x+|x|,\ |x|\le 2\pi$ is

a) 0 b) 1 c) 2 d) 3

Key. d

Sol. $x<0\Rightarrow|\cos x|=0\Rightarrow x=\frac{-\pi}{2},\frac{-3\pi}{2};\ x>0$

$\Rightarrow \cos x=2x$ has one root in $\left(0,\frac{\pi}{2}\right)$

W.E-16: Let $f:R-\left\{\frac{3}{2}\right\}\to R,\ f(x)=\frac{3x+5}{2x-3}$

Let $f_2(x)=f(f(x));f_3(x)=f(f_2(x));........,f_n(x)$ $=f\left(f_{n-1}(x)\right)$, then $f_{2008}(x)+f_{2009}(x)=$

Sol. $f(x)=\frac{3x+5}{2x-3};$

$$f_2(x)=f(f(x))=\frac{3f(x)+5}{2f(x)-3}=x$$

$$\therefore f_2(x)=f_4(x)=.....=f_{2008}(x)=x$$

$$f(x)=f_3(x)=........=f_{2009}(x)=\frac{3x+5}{2x-3}$$

$$\therefore f_{2008}(x)+f_{2009}(x)=x+\frac{3x+5}{2x-3}=\frac{2x^2+5}{2x-3}$$